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Availability: Aug. 2026 - Jul. 2027
San Francisco Bay Area

EXPERIENCE

ByteDance

Singapore

Applied Machine Learning Orchestration Intern (AML) | K8S, Ray, Go, Scheduling, Inference Mar. 2026 - Aug. 2026

- Engineered ByteDance's next-gen autonomous unified orchestrator to scale heterogeneous clusters **over 200k+ GPUs & 1M+ CPUs**, supporting Exabyte-scale training & real-time inference.
- Developing **topology-aware** placement algorithms & **JIT allocation** logic within Gödel/Kubernetes; optimising **RDMA/NCCL** communication primitives to minimise tail latency and reduce configuration overhead.
- Developing **intermediate representation** using **Task/Actor** primitives to decouple computational logic from hardware; implementing **MPMD** patterns to streamline complex multi-modal model lifecycles.
- Engineering **low-latency synchronisation protocols** for online learning pipelines, ensuring strict data consistency and model-update timeliness for TikTok's global **Search, Ads, & Recommendation** models.

Software Engineering Intern (BytePlus Platform Engineering) | Go, RAG, Next.js

Jan. 2026 - Mar. 2026

- Engineered production features for byteplus.com supporting Seedream 5.0 launch, directly impacting international conversions for **1000+ B2B clients & 440k+ MAU**.
- Leading 0→1 development of BytePlus **RAG** search: deploying **Go-driven Firecrawlers** for **ETL data ingestion**, integrated with SSE-supported **multimodal rendering** engine to stream complex LLM outputs.
- Led performance engineering initiatives, implementing **in-memory caching** & experimental **edge computing** patterns to reduce SSR reliance & slash **TTI, LCP & FCP**, enhancing UX under high-server loads.

IBM (Client Engineering) | *Spring Boot, Node.js, AWS, SQL Server, CrowdStrike, React*

Sep. 2025 - Dec. 2025

Full Stack Developer Intern

Singapore

- **Forward deployed** on-site to co-engineer **B2G** SG military personnel **RBAC** system with 65k+ DAU.
- Built **Lambda**-automated event & visitor retry management batch pipeline using **Node.js** workers, integrating **FALCON** & Sequelize for clearance, triggering notifications using **AWS SNS/SES/SQS**.
- Implemented end-to-end conditional & automated on/offboarding with real-time **cross-system validation** to handle user **role management** across multiple locations, reducing manual data entry by **80%**.

CloudJoi (Malaysia's #1 arts ticketing platform) | *Laravel, React Native, Next.js*

May. 2025 - Aug. 2025

Software Engineering Intern - cloudjoi.com

Kuala Lumpur, Malaysia

- Contributed towards AI **semantic** search & recommendations via **vector embedding** & clustering metadata.
- Launched knowledgebase.cloudjoi.com—dashboard helpdesk for **200+ clients**—0→1 on **Next.js** with RESTful API.
- Deployed **ticket check-in/out system** for all live CloudJoi shows, enabling organisers to manage attendee status.
- Integrated **password change confirmation** flow with **queue-based email notifications** & Redis caching.

PROJECTS

DrawMyRoute (AI Running Route Generator) | *RAG, Next.js, FastAPI, OSRM, Mapbox*

Jan. 2026 - Present

- Built an AI route generator for Hack&Roll 2026 that transforms text prompts into GPS-snapped running routes.
- Engineered 5-stage RAG pipeline with vector search & LLM re-ranking to match prompts to 500+ SVG library.
- Implemented multi-variant parallel optimiser testing 16 rotation/scale combinations for optimal road snapping.

Grid Universe AI Agent | *PyTorch, NumPy, Scikit-learn*

Oct. 2025 - Nov. 2025

- Engineered an autonomous agent for a stochastic grid-world, across 48 complex test levels.
- Designed a custom CNN architecture (PyTorch) to reconstruct game state from snapshots with **99% accuracy**.
- Developed a lightweight statistical cipher decoder using SVD and NumPy to interpret encrypted objectives.

EDUCATION

National University of Singapore | Bachelor of Computing in **Computer Science**

May 2027

NUS **Scholar** | Residential College Entrepreneurship Club **President** | Notion **Campus Leader** | **Teaching Assistant**

TECHNICAL SKILLS

Languages: Go, Python, Java, TypeScript, PHP, JavaScript, Swift, C, C++, SQL, R, Assembly, Bash
Frameworks: React (Native), Node.js, Next.js, Laravel, Spring Boot, Supabase, Flask, Django, FastAPI
DevOps & Tools: Kubernetes, Ray, AWS, SQL Server, MySQL, MongoDB, PostgreSQL, Git, GCP, Docker
Libraries: Matplotlib, NumPy, pandas, SwiftUI, Three.js, sklearn, PyTorch